

An IoT GAS And ENVIRONMENT MONITOR SENSOR

ASHWINI KUMAR SINHA

Environmental sensing plays a key role in managing air quality, controlling pollution, ensuring factory safety, detecting hazardous gases, and supporting laboratory monitoring. To be effective, such systems must be battery-efficient, lightweight, compact, versatile, and capable of real-time wireless data transmission. While most existing solutions are bulky, this design offers the same functions in a compact multi-gas sensor module.

The environmental sensor is designed to record gas data, humidity, and temperature alongside each gas measurement and export it as a CSV file. It includes a sleep mode option, allowing users to set intervals for the device to sleep and wake to collect data, enhancing battery efficiency. The design incorporates the BME690 (U2) sensor, which provides real-time data on humidity, air pressure, temperature, VOC (volatile organic compound) gas resistance, and an estimated indoor air quality (IAQ) index. While the IAQ is not highly accurate, it meets most environmental monitoring requirements. Additionally, the MICS-6814 sensor is integrated to measure gases such as CO, NO₂, NH₃, C₃H₈, C₄H₁₀, CH₄, H₂, and C₂H₅OH.

For IoT functionality and data processing, a small but powerful development board, the IndusBoard Coin, is used. An OLED display can be added to display basic environmental data in real time. The FPC (flexible printed circuit) sensor module is designed for seamless integration with the IndusBoard Coin ecosystem, making it compact,

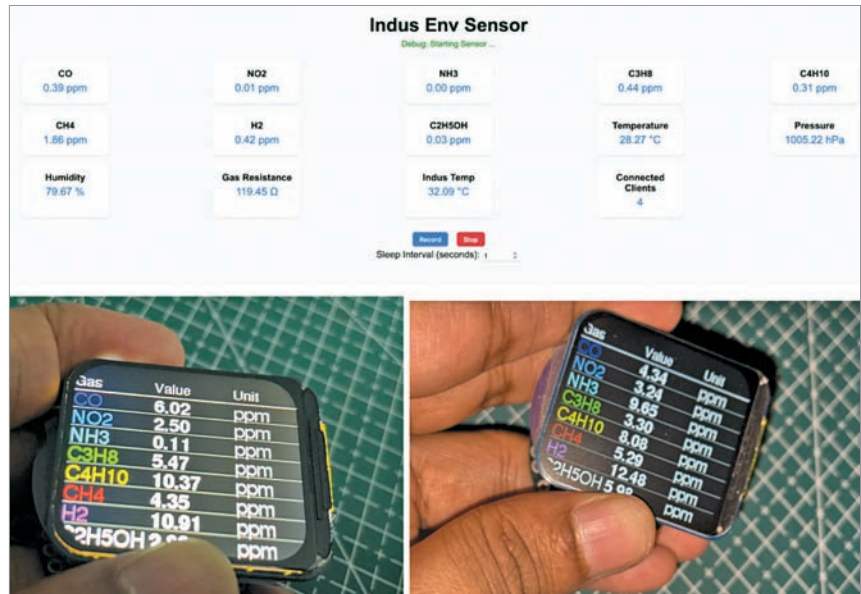


Fig. 1: IoT dashboard of sensor data on the author's prototype

BILL OF MATERIALS

Components	Name	Quantity
IndusBoard Coin V2	U1	1
BME 690	U2	1
MICS 6814	U3	1
GC9A01/LCD	LCD1	1
24-pin 0.5mm FPC ribbon cable		1
Battery 3.3V-3.7V		1

reliable, and easy to use. Fig. 1 shows the IoT dashboard of sensor data on the prototype. To build this device, the required components are listed in the Bill of Materials table.

Software

To begin, install the necessary libraries for the sensors and display. All library files and code can be downloaded by scanning the QR code at the bottom of this article. Copy the libraries to the Arduino library folder. In the code, configure

```

IoT_Sensor.ino  Sensor_ran.ino
1  #include <WiFi.h>
2  #include <ESPAsyncWebServer.h>
3  #include <AsyncTCP.h>
4  #include <Wire.h>
5  #include <MICS6814-I2C.h>
6  #include "bme69x.h"
7  #include <esp_sleep.h>
8  #include <NTPClient.h>
9  #include <WiFiUdp.h>
10 #include <SPI.h>
11 #include <TFT_eSPI.h>
12 #include <Wire.h>
13 // WiFi AP credentials
14 const char* ssid = "IndusEnvSensor";
15 const char* password = "password123";
16

```

Fig. 2: Code snippet configuring Wi-Fi SSID and password

the Wi-Fi SSID and password as required. By default, the Wi-Fi is set to access point (AP) mode, but it can be configured to station (STA) mode to connect to a home or office router, or both modes can be enabled by adjusting the code accordingly. Sensor data is captured and updated on the attached display